

2024 Paper 2 Questions by Topic

- 1 (a) State the principle of conservation of momentum.

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..... [2]

- (b) An object of mass $2m$ is travelling at a speed of 5.0 m s^{-1} in a straight line. It collides with an object of mass $3m$ which is initially stationary, as shown in Fig. 3.1.

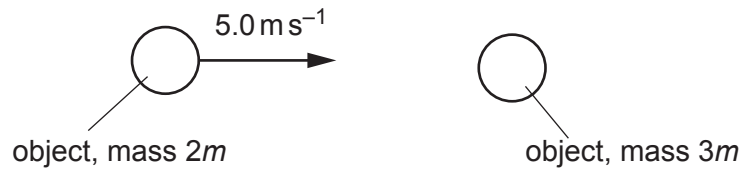


Fig. 3.1

After the collision, the object of mass $2m$ moves with velocity v at an angle of 30° to its original direction of motion.

The object of mass $3m$ moves with velocity w also at an angle of 30° , as shown in Fig. 3.2.

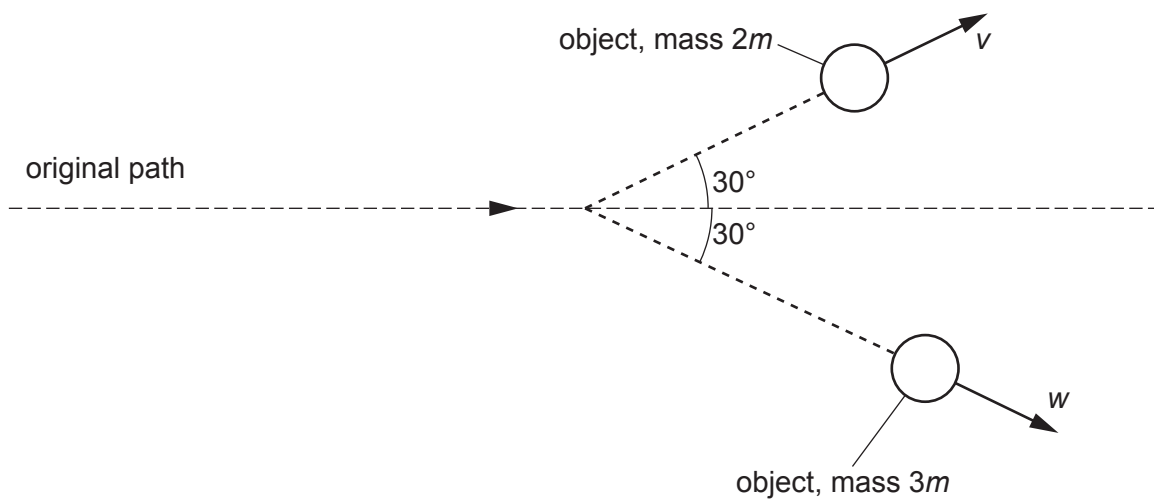


Fig. 3.2

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By considering the conservation of momentum in two dimensions, calculate the magnitudes of v and w .

$$v = \dots\dots\dots \text{ms}^{-1}$$

$$w = \dots\dots\dots \text{ms}^{-1}$$

[4]

- (c) An object of mass 4.2 kg is travelling in a straight line at a speed of 6.0 ms^{-1} . The object is brought to rest in a distance of 0.050 m by a constant force.

Calculate the magnitude of this force.

$$\text{force} = \dots\dots\dots \text{N} \quad [3]$$

[Total: 9]

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- 2 (a) Define linear momentum.

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 [1]

- (b) A car of mass 1800 kg is moving in a straight line. Fig. 2.1 shows the variation with time t of the momentum p of the car.

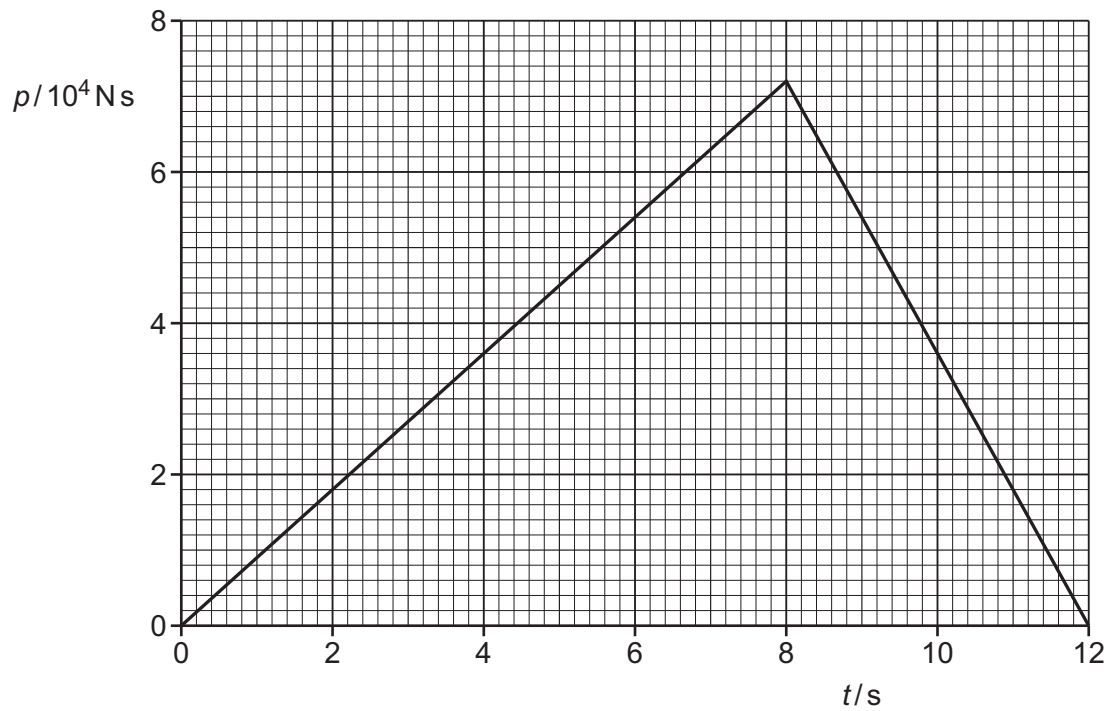


Fig. 2.1

- (i) Calculate the maximum speed reached by the car.

maximum speed = ms^{-1} [1]

- (ii) Calculate the maximum kinetic energy of the car.

maximum kinetic energy = J [2]

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- (iii) Show that the acceleration of the car at time $t = 4.0\text{ s}$ is 5.0 m s^{-2} .

[2]

- (iv) Determine the distance travelled by the car between times $t = 0$ and $t = 12.0\text{ s}$.

distance = m [2]

- (c) On Fig. 2.2, sketch the variation with time t of the acceleration a of the car in (b) from $t = 0$ to $t = 12.0\text{ s}$.

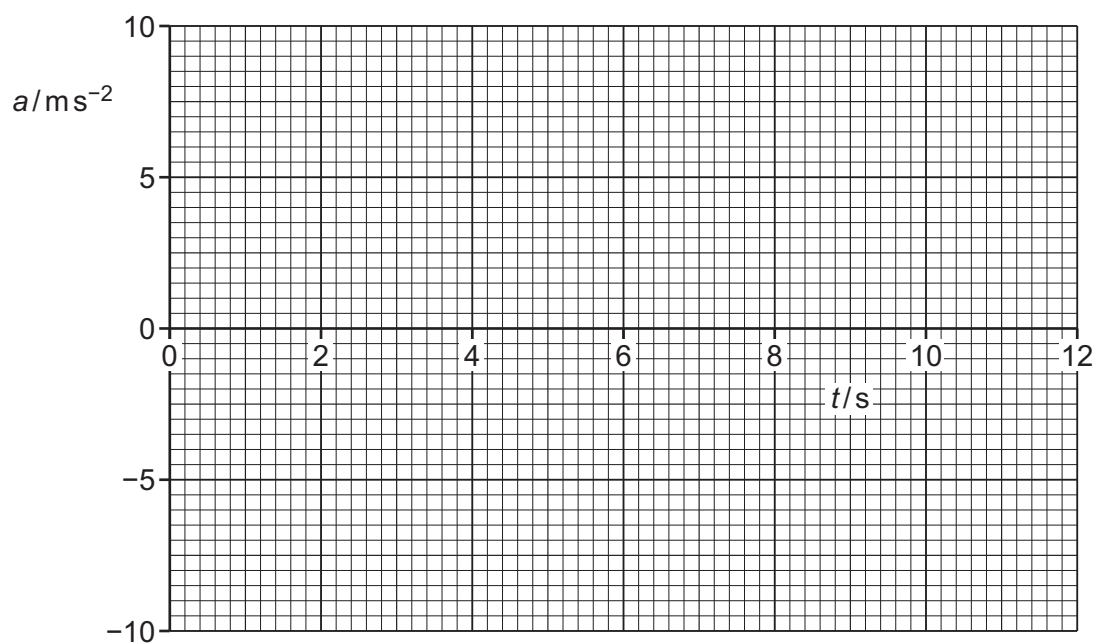


Fig. 2.2

[3]

[Total: 11]

- 2 (a) Define momentum.

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 [1]

- (b) A child stands on a scooter on horizontal ground. The combined mass of the child and the scooter is 16 kg.
 The child starts from rest and pushes once on the ground with her foot which causes her to accelerate. The push lasts for a time of 1.1 s. The speed of the child and the scooter after the push is 0.60 m s^{-1} .

Determine the average resultant force acting horizontally on the child and the scooter during the push.

average force = N [2]

- (c) Later, the child in (b) travels down a slope at a constant angle to the horizontal, as shown in Fig. 2.1.

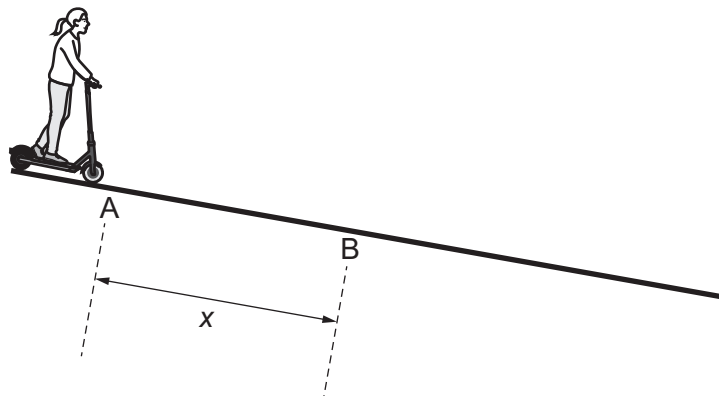


Fig. 2.1 (not to scale)

At point A her speed is 0.60 m s^{-1} . She has a constant acceleration of 0.85 m s^{-2} parallel to the slope. After a time of 3.7 s, she reaches point B.

Calculate the distance x travelled by the child along the slope from A to B.

$x = \dots\dots\dots \text{ m}$ [2]

- (d) At point B, the child in (c) applies the brake with a constant force to maintain a constant velocity. Point C is 18 m from point B, as shown in Fig. 2.2.

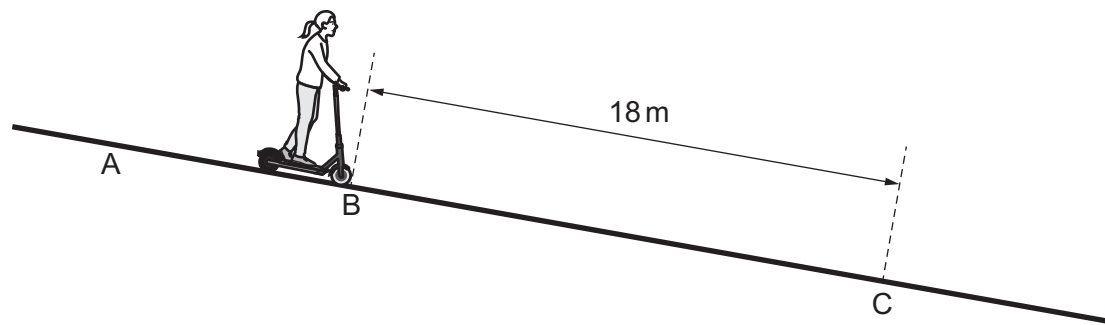


Fig. 2.2 (not to scale)

The work done by the braking force between B and C is 250 J.

- (i) Determine the magnitude of the braking force.

force = N [2]

- (ii) On Fig. 2.3, sketch the variation of the kinetic energy of the child and scooter with distance travelled from point A to point C. Numerical values for kinetic energy are not required.

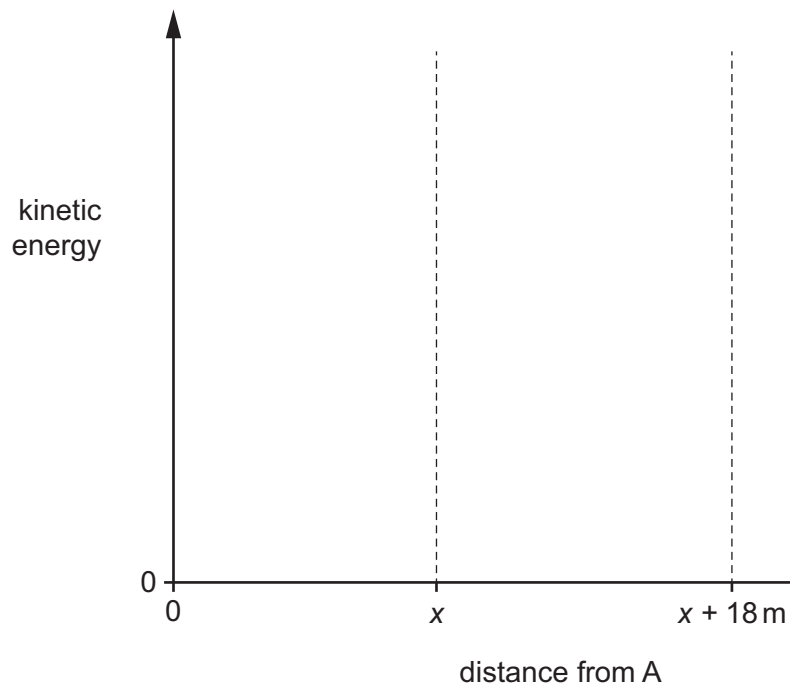


Fig. 2.3

[3]

[Total: 10]

- 3 (a) State the principle of conservation of momentum.

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..... [2]

- (b) A ball X has mass 240 g and moves in a straight line on a horizontal frictionless surface with an initial speed of 16 m s^{-1} . The ball collides with a stationary ball Y that has mass 480 g. After the collision, ball X is stationary, as shown in Fig. 2.1.

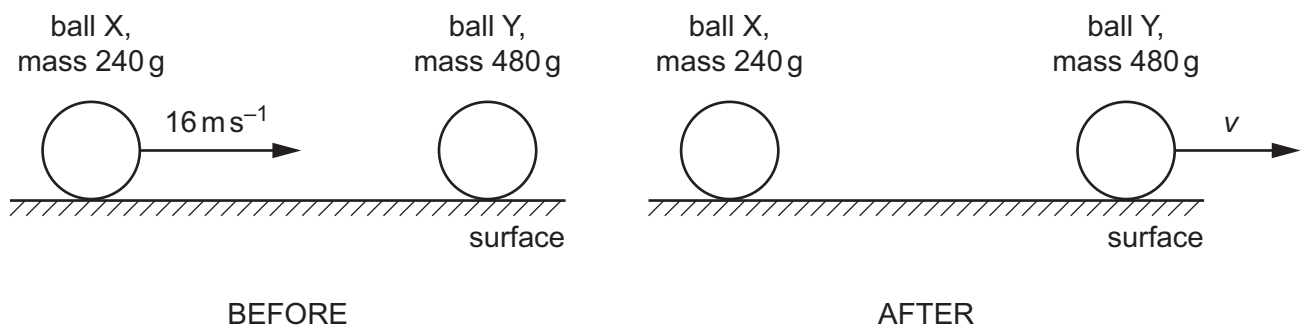


Fig. 2.1

- (i) Show that the speed v of ball Y after the collision is 8.0 m s^{-1} .

[1]

- (ii) Calculate the change in the total kinetic energy ΔE_K of the balls due to the collision.

$\Delta E_K = \dots\dots\dots \text{ J [3]}$

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- (c) The collision in (b) lasts for a time of 2.0 ms. Assume that the contact force between the balls is constant during this time.
- (i) Determine the magnitude and direction of the force exerted on ball X by ball Y during the collision.

magnitude = N

direction
[3]

- (ii) Compare the magnitude and direction of the force exerted on ball Y by ball X during the collision with the answers in (c)(i). No further calculations are required.

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..... [2]

[Total: 11]